PRACTICE TASK 3:

Members::

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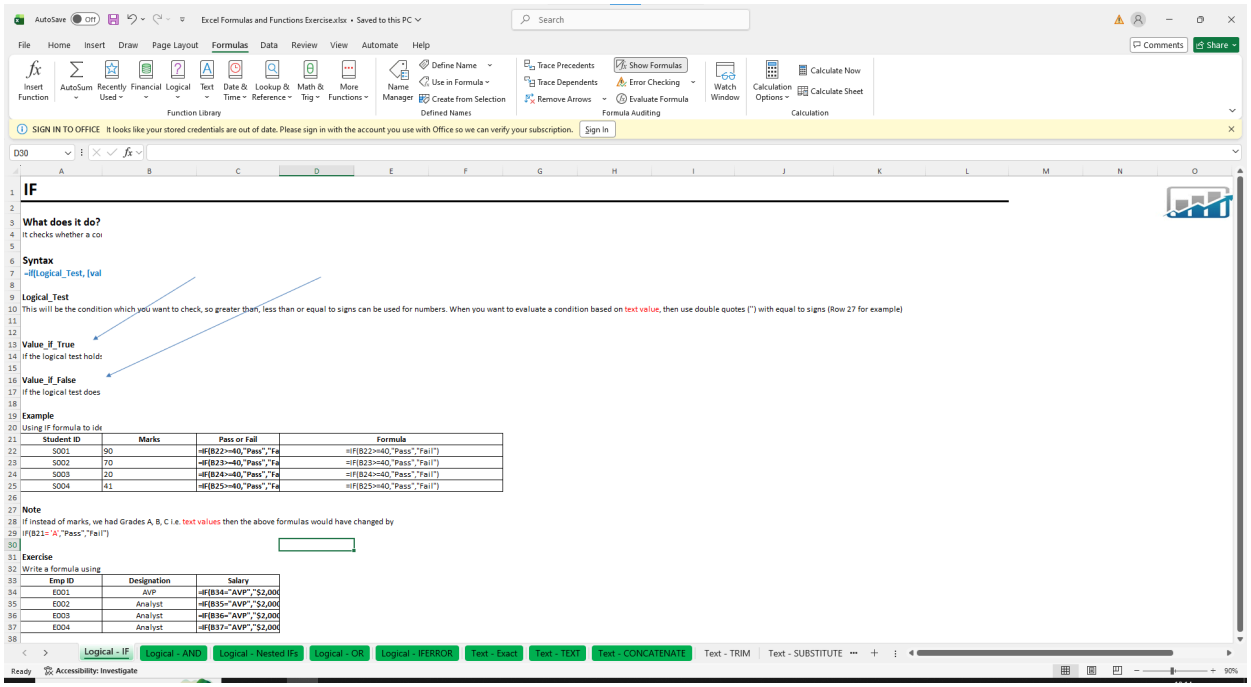
1. IF FUNCTION



IF

What does it do?

It checks whether a co

Syntax

=IF(Logical_Test, [val

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used for numbers. When you want to evaluate a condition based on text value, then use double quotes ("") with equal to signs (Row 27 for example)

Value_If_True

If the logical test holds

Value_If_False

If the logical test does

Example

Using IF formula to ide

Student ID	Marks	Pass or Fail	Formula
S001	90	=IF(B22>=40,"Pass","Fa	=IF(B22>=40,"Pass","Fail")
S002	70	=IF(B23>=40,"Pass","Fa	=IF(B23>=40,"Pass","Fail")
S003	20	=IF(B24>=40,"Pass","Fa	=IF(B24>=40,"Pass","Fail")
S004	41	=IF(B25>=40,"Pass","Fa	=IF(B25>=40,"Pass","Fail")

Note

If instead of marks, we had Grades A, B, C i.e. text values then the above formulas would have changed by

IF(B21='A',"Pass","Fail")

Exercise

Write a formula using

Emp ID	Designation	Salary
E001	AVP	=IF(B34="AVP","$2,000
E002	Analyst	=IF(B35="AVP","$2,000
E003	Analyst	=IF(B36="AVP","$2,000
E004	Analyst	=IF(B37="AVP","$2,000

2. AND FUNCTION

AND

What does it do?

Checks if all arguments are true and returns TRUE value else returns FALSE value. You need to have minimum 2 conditions to check. AND function is used with IF condition a lot

Syntax

=AND(Logical 1, Logica

Logical 1

First condition which ne

Logical 2

Second condition whic

Example 1

Testing condition 1 and

Condition 1	Condition 2	Result	Formula
1	2	=AND(A19>0,B19>0)	=AND(A19>0,B19>0)
1	-2	=AND(A20>0,B20>0	=AND(A20>0,B20>0)

Example 2

Use IF with AND function to check if student has passed or not. If the student has failed in 1 of the 2 subjects, then the student will be considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	=IF(AND(B26="PASS",C26="PA	=IF(AND(B26="PASS",C26="PASS"),"PASS","FAIL")
S002	PASS	FAIL	=IF(AND(B27="PASS",C27="PA	=IF(AND(B27="PASS",C27="PASS"),"PASS","FAIL")
S003	FAIL	PASS	=IF(AND(B28="PASS",C28="PA	=IF(AND(B28="PASS",C28="PASS"),"PASS","FAIL")
S004	PASS	PASS	=IF(AND(B29="PASS",C29="PA	=IF(AND(B29="PASS",C29="PASS"),"PASS","FAIL")

Exercise

Write a formula using AND with IF to find if an employee should get a "GOOD" increment or not. Sales made by the employee should be more than 50 and he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)
E001	AVP	51	YES	=IF(AND(D36="YES",C36>50),"Y
E002	Analyst	100	NO	=IF(AND(D37="YES",C37>50),"Y
E003	Analyst	49	YES	=IF(AND(D38="YES",C38>50),"Y
E004	Analyst	75	YES	=IF(AND(D39="YES",C39>50),"Y

3. NESTED IF

Nested IF

From Excel 2007 onwards,

Example

Use Nested IF to put the gr

TABLE A	
Score	Grade
> 70	A
41-70	B
< 40	C

TABLE B			
Student ID	Marks	Grade	Formula
S001	90	=IF(E26< 41,"C",IF(E26< 71,"	=IF(E26< 41,"C",IF(E26< 71,"B","A"))
S002	70	=IF(E27< 41,"C",IF(E27< 71,"	=IF(E27< 41,"C",IF(E27< 71,"B","A"))
S003	20	=IF(E28< 41,"C",IF(E28< 71,"	=IF(E28< 41,"C",IF(E28< 71,"B","A"))
S004	41	=IF(E29< 41,"C",IF(E29< 71,"	=IF(E29< 41,"C",IF(E29< 71,"B","A"))

Exercise

Write a formula using Nested IF conditions to check what level increment an employee should get and then convert the amount into dollars using Format

TABLE A	
Performance	Increment
A	1000
B	500
C	250

TABLE B			
Emp ID	Designation	Performance	Result
E001	AVP	A	=IF(F36="A",B36,I
E002	Analyst	A	=IF(F37="A",B36,I
E003	Analyst	B	=IF(F38="A",B36,I
E004	Analyst	C	=IF(F39="A",B36,I

4. OR FUNCTION

OR

What does it do?

Checks if any of the conditions are true or not. It returns TRUE value even if one of the conditions holds else returns FALSE value. Basically, for a FALSE to appear, no conditions should hold. You need to have minimum 2 conditions to check. AND function is used with IF condition a lot

Syntax

=OR(Logical 1, Logical 2

Logical 1

First condition which ne

Logical 2

Second condition whic

Example 1

Testing condition 1 and

Condition 1	Condition 2	Result	Formula
1	2	=OR(A20>0,B20>0)	=OR(A20>0,B20>0)
1	-2	=OR(A21>0,B21>0)	=OR(A21>0,B21>0)
-1	-2	=OR(A22>0,B22>0)	=OR(A22>0,B22>0)

Example 2

Use IF with OR function to check if student has passed or not. If the student has failed in 1 of the 2 subjects, then the student will be considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	=IF(OR(B28="FAIL",C28="FAIL"	=IF(OR(B28="FAIL",C28="FAIL"),"FAIL","PASS")
S002	PASS	FAIL	=IF(OR(B29="FAIL",C29="FAIL"	=IF(OR(B29="FAIL",C29="FAIL"),"FAIL","PASS")
S003	FAIL	PASS	=IF(OR(B30="FAIL",C30="FAIL"	=IF(OR(B30="FAIL",C30="FAIL"),"FAIL","PASS")
S004	PASS	PASS	=IF(OR(B31="FAIL",C31="FAIL"),	=IF(OR(B31="FAIL",C31="FAIL"),"FAIL","PASS")

Exercise

Write a formula using OR with IF to find if an employee should get a "GOOD" increment or not. Either Sales made by the employee should be more than 50 or he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)
E001	AVP	51	YES	=IF(OR(C38>50,D38="YES"),"YE
E002	Analyst	49	NO	=IF(OR(C39>50,D39="YES"),"YE
E003	Analyst	49	YES	=IF(OR(C40>50,D40="YES"),"YE
E004	Analyst	75	YES	=IF(OR(C41>50,D41="YES"),"YES

5. IFERROR FUNCTION

IFERROR

What does it do?
Sometimes when we use formula in cells, we might get an error because of the cell value, IFERROR allows us to remove such errors with values we want

Syntax
=IFERROR(Value,[Valu

Logical 1
First cell value which is

Value_if_error
If in the cell there is an e

Example 1
Divide Col A by Col B ar

Col A	Col B	Without IFERROR &Formula		With IFERROR &Formula	
1	2	=A19/B19	=A19/B19	=IFERROR(A19/B19,999)	=IFERROR(A19/B19,999)
1	0	=A20/B20	=A20/B20	=IFERROR(A20/B20,999)	=IFERROR(A20/B20,999)

Note
In the above example, on dividing 1 by 0, we get an error, without using IFERROR we get #DIV/0! i.e. an error. So using IFERROR, we can give values we want. You can also enter text which can let the user know that there is some problem with the data.

Example 2
Divide Col A by Col B ar

Col A	Col B	Without IFERROR &Formula		With IFERROR &Formula	
1	2	=A29*B29	=A29*B29	=IFERROR(A29*B29,"DATA IS W	=IFERROR(A29*B29,"DATA IS WRONG")
1	1	=A30*B30	=A30*B30	=IFERROR(A30*B30,"DATA IS W	=IFERROR(A30*B30,"DATA IS WRONG")
1	a	=A31*B31	=A31*B31	=IFERROR(A31*B31,"DATA I	=IFERROR(A31*B31,"DATA IS WRONG")
5	a	=A32*B32	=A32*B32	=IFERROR(A32*B32,"DATA	=IFERROR(A32*B32,"DATA IS WRONG")

Exercise
Write a formula using IFERROR. Find the monthly salary of each employee using data in Column C and if an error appears, then change it to ERROR IN SALARY. Change the format to £ (Pounds) and no decimal places

Emp ID	Designation	Annual Salary	Monthly Salary
E001	AVP	80000	=IFERROR(C39/12,"ERROR
E002	Analyst	50000	=IFERROR(C40/12,"ERROR
E003	Analyst	abcd	=IFERROR(C41/12,"ERROR
E004	Analyst	40000	=IFERROR(C42/12,"ERROR

6. EXACT FUNCTION

EXACT

What does it do?
It checks whether 2 values (texts, numbers) are exactly same or not. If they are same, then it will return TRUE else it will return FALSE. This function is case-sensitive

Syntax
=EXACT(Text1, Text2)

Text1
First string which you wa

Text2
Second string which you

Example 1
Testing condition 1 and c

Col A	Col B	Result	Formula
EXCEL	EXCEL	=EXACT(A19,B19)	=EXACT(A19,B19)
EXCEL	Excel	=EXACT(A20,B20)	=EXACT(A20,B20)

Note
In the above example, Row 20 result is FALSE because the EXACT function is case-sensitive. The first value is in upper capse (EXCEL) whereas the second one (Excel) is a combination of upper and lower capse

Exercise
Write a formula using EXACT to check if the values in Col A and Col B are same or not

Col A	Col B	Results
TEST	Test	=EXACT(A29,B29)
TEST	TEST	=EXACT(A30,B30)
1234	1234	=EXACT(A31,B31)
1.1	1.101	=EXACT(A32,B32)

7. TEXT FUNCTION

TEXT

What does it do?

It converst a numeric value into text based on the format given by the user

Syntax

=TEXT(Value, Format_te

Value

Value for which we need t

Format_text

Format specified by the us

Example 1

Using different text format

Col A	Result	Formula
42821	=TEXT(A18, "dd/mm/yyyy")	=TEXT(A18, "dd/mm/yyyy")
42821	=TEXT(A19, "dd/m/yyyy")	=TEXT(A19, "dd/m/yyyy")
42821	=TEXT(A20, "mmm dd yyyy")	=TEXT(A20, "mmm dd yyyy")
42821	=TEXT(A21,"mmm-yy")	=TEXT(A21,"mmm-yy")
14.141414	=TEXT(A22, "0.00")	=TEXT(A22, "0.00")
14598.12	=TEXT(A23, "$#,##0.00")	=TEXT(A23, "$#,##0.00")
14598.12	=TEXT(A24, "£#,##0.00")	=TEXT(A24, "£#,##0.00")
0.583333333333333	=TEXT(A25, "hh:mm")	=TEXT(A25, "hh:mm")

Note:

Col A Row 19 & 20 - Excel stores dates in numbers, so that is why when we use Text with Date formats, Excel converts those numbers dates

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Value	Convert To	Results
42876	42876	**=EXACT(A34,B34)**
42876	42876	**=EXACT(A35,B35)**
0.5	0.5	**=EXACT(A36,B36)**
12345.78	12345.78	**=EXACT(A37,B37)**
12345.78	12346	**=EXACT(A38,B38)**

8. CONCATENATE FUNCTION

CONCATENATE

What does it do?
This function joins se

Syntax
=CONCATENATE(Stri
OR
=String1&String2& s

String 1
First string which you

String 2
Second string which

Example
Join the 3 strings tog

String1	String2	String3	Result	Formula
S001	90	Pass	**=CONCATENATE(A20,B20,**	=CONCATENATE(A20,B20,C20)
S002	70	Pass	**=CONCATENATE(A21,B21,C**	=CONCATENATE(A21,B21,C21)
S003	20	Fail	**=A22&B22&C22**	=A22&B22&C22
S004	41	Pass	**=A23&B23&C23**	=A23&B23&C23
S005	39	Fail	**=CONCATENATE(A24," ",B2**	=CONCATENATE(A24," ",B24," ",C24)
S006	99	Pass	**=A25&" "&B25&" "&C25**	=A25&" "&B25&" "&C25
S007	11	Fail	**=CONCATENATE(A26,":",B**	=CONCATENATE(A26,":",B26,":",C26)

Note
In order to add spaces, other special characters between strings, you would need to use double quotes "" and declare the value between them as done in the examples above

Exercise
Join the 3 strings tog

String1	String2	String3	Result	USE	Formula
E001	AVP	800	**E001-AVP-800**	CONCATENATE	**=CONCATENATE(A35,"-",B35,"-",C35)**
E002	Analyst	500	**E002Analyst500**	CONCATENATE	**=CONCATENATE(A36,B36,C36)**
E003	Analyst	100	**E003/Analyst-100**	CONCATENATE	**=CONCATENATE(A37,"/",B37,"-",C37)**
E004	Analyst	400	**E004:Analyst:400**	&	**=A38&":"&B38&":"&C38**
E005	AVP	1000	**E005 AVP 1000**	&	**=A39&" "&B39&" "&C39**
E006	Analyst	250	**E006 Analyst 250**	&	**=A40&" "&B40&" "&C40**
E007	Analyst	500	**E007;Analyst"500**	&	**=A41&";"&B41&""&C41**

9. EVEN FUNCTION

EVEN

What does it do?

This function rounds a positive number up and negative number down to the nearest even integer

Syntax

=EVEN(Number/Cell)

Text

Cell which you want to conve

Example

Use EVEN function on the foll

Numbers	Rounded	Formula
45.3976105679829	=EVEN(A15)	=EVEN(A15)
-50.6546264870099	=EVEN(A16)	=EVEN(A16)
85.4139092599847	=EVEN(A17)	=EVEN(A17)
31.5893471217502	=EVEN(A18)	=EVEN(A18)

Exercise

Use EVEN function and round

Items	Result
1111.111	=EVEN(A23)
95.3	=EVEN(A24)
51.5	=EVEN(A25)
-4.6985	=EVEN(A26)

10. MOD FUNCTION

MOD

What does it do?

This function returns the remainder after a number is divided by a divisor

Syntax

=MOD(number,divisor)

Number

This is the number which you

Divisor

The number to divide with

Example

Use MOD function on the foll

Number	Divisor	Remainder	Formula
3	5	=MOD(A18,B18)	=MOD(A18,B18)
19	3	=MOD(A19,B19)	=MOD(A19,B19)
8	2	=MOD(A20,B20)	=MOD(A20,B20)
-8.5	3	=MOD(A21,B21)	=MOD(A21,B21)

Exercise

Use MOD function to find the

Number 1	Number 2	Result
5	6	=MOD(A26,B26)
9	2	=MOD(A27,B27)
8	5	=MOD(A28,B28)
-4.6985	-1	=MOD(A29,B29)

11. AVERAGE FUNCTION

AVERAGE

What does it do?

This function returns the average value of the given numbers - It ignores text values and blank cells

Syntax

=AVERAGE(Number1,Num

Number1,Number2 and so

List of values from which you

Example

Use AVERAGE function to find

Number1	Number2	Number3	Result	Formula
3		5	=AVERAGE(A15:C15)	=MAX(A15:C15)
19	a	1	=AVERAGE(A16:C16)	=MAX(A16:C16)
8	b	c	=AVERAGE(A17:C17)	=MAX(A17:C17)
-8.5	3	0.5	=AVERAGE(A18:C18)	=MAX(A18:C18)

Exercise

Calculate AVERAGE for the fo

Number1	Number2	Number3	Result
3	5	3	=AVERAGE(A23:C23)
19	a	1	=AVERAGE(A24:C24)
8	b	c	=AVERAGE(A25:C25)
1.5		4.5	=AVERAGE(A26:C26)

12. SMALL FUNCTION

SMALL

What does it do?

This function returns the nth smallest number from the given set of values

Syntax

=SMALL(Array, nth small

Array

These are list of numbers fro

nth smallest number

The nth smallest number wh

Example

Use SMALL function to find th

Array/List of Numbers			nth smallest number	Result	Formula
3	1	5	1	=SMALL(A18:C21,D18)	=SMALL(A18:C21,D18)
19	50	1	5	=SMALL(A18:C21,D19)	=SMALL(A18:C21,D19)
8	12	20	4	=SMALL(A18:C21,D20)	=SMALL(A18:C21,D20)
-8.5	3	0.5	12	=SMALL(A18:C21,D21)	=SMALL(A18:C21,D21)

Exercise

Calculate nth smallest numb

Array/List of Numbers			nth smallest number	Result
3	5	3	1	=SMALL(A26:C29,D26)
19	2	1	2	=SMALL(A26:C29,D27)
8	5	3	3	=SMALL(A26:C29,D28)
1.5	1	4.5	4	=SMALL(A26:C29,D29)

13. LARGE FUNCTION

LARGE

What does it do?

This function returns the nth largest number from the given set of values

Syntax

=LARGE(Array, nth larges

Array

These are list of numbers fro

nth largest number

The nth largest number whic

Example

Use SMALL function to find th

Array/List of Numbers			nth largest number	Result	Formula
3	1	5	1	=LARGE(A18:C21,D18)	=LARGE(A18:C21,D18)
19	50	1	5	=LARGE(A18:C21,D19)	=LARGE(A18:C21,D19)
8	12	20	4	=LARGE(A18:C21,D20)	=LARGE(A18:C21,D20)
-8.5	3	0.5	12	=LARGE(A18:C21,D21)	=LARGE(A18:C21,D21)

Exercise

Calculate nth largest number

Array/List of Numbers			nth largest number	Result
3	5	3	1	=LARGE(A26:C29,D26)
19	2	1	2	=LARGE(A26:C29,D27)
8	5	3	3	=LARGE(A26:C29,D28)
1.5	1	4.5	4	=LARGE(A26:C29,D29)

14. COUNT IF FUNCTION

COUNTIF

What does it do?

This function returns the count of number of cells which consist of numbers and meet a given condition

Syntax

=COUNTIF(Range,Criteria)

Range

Range of cells which you want to count

Criteria

Condition which you want to check and then count the cells on the basis of it

Example

Use COUNTIF function on Col A, B, C and D to find the number of numeric values, based on values being greater or less as specified in Col E

Col A	Col B	Col C	Col D	Col E	Result	Formula
3	1	5	3	Values > 1	3	=COUNTIF(A18:D18,">1")
19	A		0	Values = 0	1	=COUNTIF(A19:D19,"=0")
8	D	20	2	Values >= 2	3	=COUNTIF(A20:D20,">=2")
-8.5	0	B	-1	Values < 0	2	=COUNTIF(A21:D21,"<0")

Exercise

Use COUNTIF function on Col A, B, C and D to find the number of numeric values, based on values being greater or less as specified in Col E

Col A	Col B	Col C	Col D	Col E	Result
1	-1	5	3	Values > 2	2
19	A		0	Values >= 0	2
8	-50	20	2	Values >= 20	1
-8.5	0		-1	Values <= 0	3